

Cardiosense: Human Activity Recognition Using ECG Signals and Hybrid Machine Learning Model

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Abstract—In recent years, the classification of human physical activities using physiological signals has become an important research area with vast applications in healthcare monitoring, fitness tracking, rehabilitation systems, and assisted living. Among various bio signals, Electrocardiogram (ECG) data has shown promising potential in recognizing different forms of human activity, as it provides insights into cardiac responses under physical exertion. This paper introduces a machine learning-based web application designed to classify human physical activities based on ECG signals. The system employs a hybrid classification model, combining the strengths of multiple algorithms to improve prediction accuracy and robustness. Trained on a well-curated dataset of ECG signals labelled with 12 distinct activities, the model predicts the most likely physical activity from unseen ECG data. Developed using Flask with a user-friendly HTML/CSS interface, the application allows CSV uploads, processes the data, and returns activity predictions. The system supports packaging for offline use and deployment on cloud platforms. This work demonstrates the potential of ECG signals in activity recognition and lays the groundwork for real-time, scalable health monitoring applications.

Keywords—*Electrocardiogram (ECG), Human Activity Recognition (HAR), Hybrid Model, Heart Beat, Physiological Signals.*

I. BACKGROUND

Human Activity Recognition (HAR) is an emerging interdisciplinary field that intersects machine learning, wearable technology, and biomedical signal processing. It focuses on the automatic identification of physical activities performed by individuals using data collected from sensors. Traditionally, HAR has relied heavily on inertial sensors like accelerometers and gyroscopes, which are integrated into smartphones, fitness trackers, and smartwatches. These devices are effective in capturing movement patterns but may lack deeper physiological insight.

Electrocardiogram (ECG) signals, on the other hand, have primarily been used in clinical settings for monitoring heart health and diagnosing cardiovascular conditions. However, recent research has revealed that ECG signals also reflect significant changes corresponding to different physical activities. This includes variations in heart rate, signal morphology, and rhythm induced by motion and exertion. These physiological changes present a unique and promising opportunity to use ECG data for recognizing and classifying human physical activities, especially in environments where other sensors may not be feasible or accurate.

The motivation behind this project stems from the growing integration of ECG monitoring in wearable devices, such as smartwatches and portable ECG patches. These advancements create a timely opportunity to expand the utility of ECG data

beyond heart health monitoring into activity recognition, rehabilitation tracking, and elderly care systems.

Despite its potential, HAR using ECG faces several challenges: noise and artifacts in the signal due to movement, variability in ECG morphology among individuals, and the need for sophisticated preprocessing and modeling techniques. To address these issues, this project proposes a robust and scalable machine learning pipeline that uses a hybrid model approach. The system aims to not only improve recognition accuracy but also offer a practical deployment through a user-friendly web application.

In essence, this project explores the novel idea of repurposing ECG signals for human activity classification, contributing to the field by offering a working prototype that combines advanced signal processing, machine learning, and web development into a single, deployable application.

II. LITERATURE REVIEW

The integration of electrocardiogram (ECG) signals in human activity recognition (HAR) has garnered significant attention in recent research, leading to the development of innovative methodologies that enhance the accuracy and applicability of HAR systems. Two notable studies exemplify these advancements:

A. "DT4ECG: A Dual-Task Learning Framework for ECG-Based Human Identity Recognition and Human Activity Detection"

This study introduces DT4ECG, an innovative dual-task learning framework designed for simultaneous human identity recognition and activity detection using ECG signals. The framework employs a robust one-dimensional convolutional neural network (1D-CNN) integrated with residual blocks to extract discriminative ECG features. A novel Sequence Channel Attention (SCA) mechanism is proposed, combining channel-wise and sequential context attention to prioritize informative features across temporal and channel dimensions. To address gradient imbalance in multi-task learning, the study integrates GradNorm, dynamically adjusting loss weights based on gradient magnitudes to ensure balanced training across tasks. Experimental results demonstrate superior performance, achieving accuracy rates of 99.12% in identity classification and 90.11% in activity classification. These findings underscore the potential of the DT4ECG framework in enhancing security and user experience across various applications, such as fitness monitoring and personalized healthcare, presenting a transformative approach to integrating ECG-based biometrics in everyday technologies.

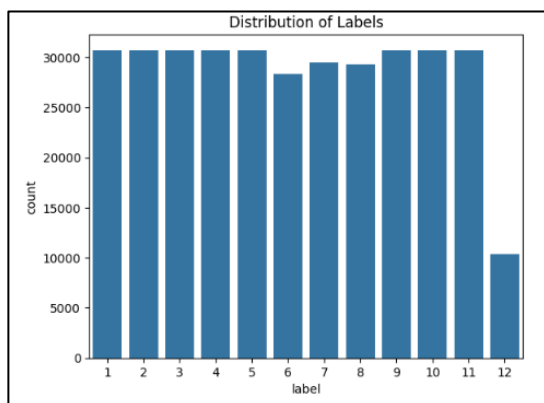
B. "Clinical Human Activity Recognition Based on a Wearable Patch of ECG and Accelerometer Sensors"

This study examines human behavioral activities by integrating acceleration signals and ECG signals collected during various activities from 20 healthy individuals and 25 patients with pneumoconiosis, using a wearable patch. A deep learning model combining Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) networks is proposed for automatic feature extraction in HAR. Spatial features and spatiotemporal features are extracted by CNN and LSTM, respectively, and the impact of the ECG signal on HAR is analyzed. The study introduces physiological signaling ECG to HAR and assesses it using several deep models, demonstrating that the inclusion of ECG signals can enhance the accuracy of activity recognition systems. The main contributions include the integration of ECG signals into HAR, the development of a CNN–LSTM model for feature extraction, and the analysis of ECG's impact on activity recognition.

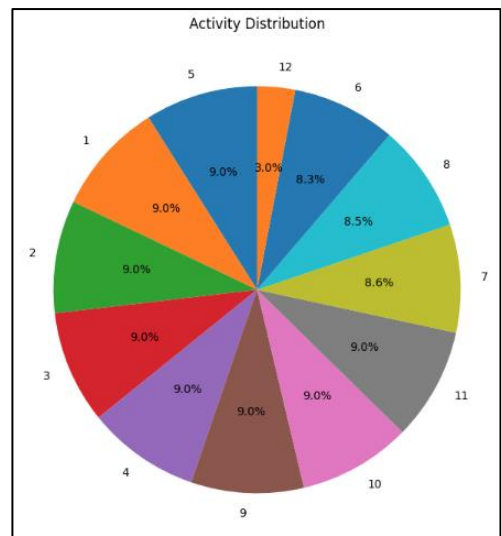
These studies highlight the evolving landscape of HAR, emphasizing the pivotal role of ECG signals and advanced machine learning models in enhancing activity recognition accuracy and reliability. The integration of ECG data not only enriches the feature set for HAR but also opens new avenues for applications in healthcare and personalized monitoring systems.

III. IMPLEMENTATION

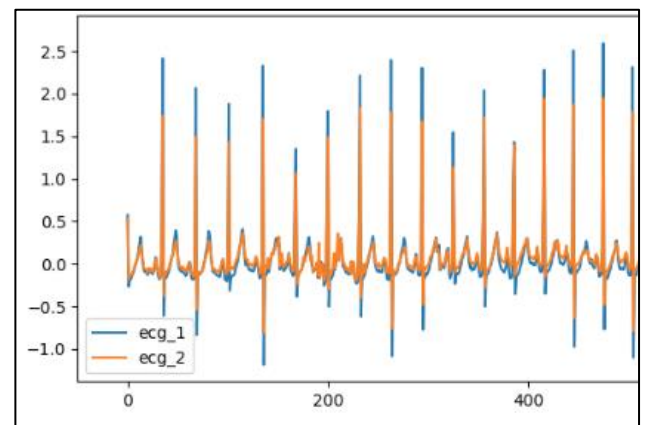
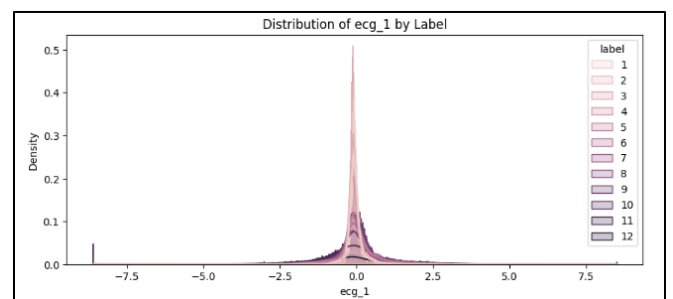
The implementation of the Cardiosense project was carried out in a structured multi-phase approach, combining data preparation, model training, evaluation, and web application development. The first step involved preprocessing the MHEALTH dataset, which contained raw ECG signals corresponding to 12 different physical activities. Using `Data_preparation.ipynb`, irrelevant columns were removed, and the data was cleaned to ensure quality inputs. Label encoding was applied to convert activity labels into numerical values, followed by normalization using `StandardScaler` to standardize the features for consistent model performance.



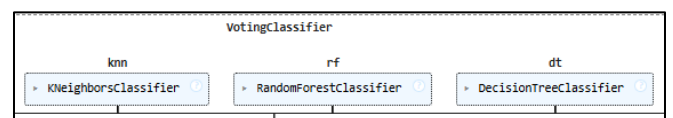
Exploratory Data Analysis (EDA) in the `EDA.ipynb` notebook helped to visualize patterns and distribution of ECG signals among activities.



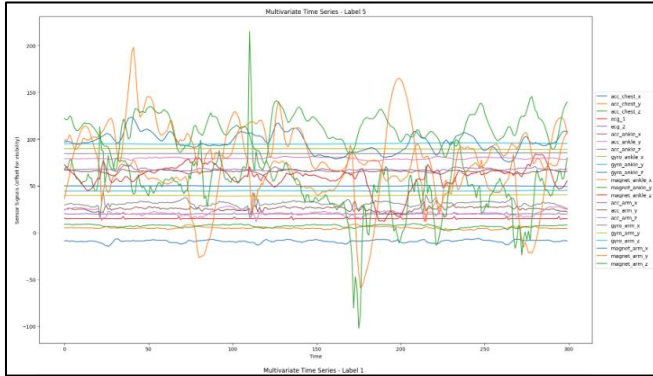
Statistical plots and correlation heatmaps were used to better understand the relationship between features and class labels, which informed the feature selection strategy.



The model development, handled in `Model.ipynb`, focused on building a hybrid ensemble classifier by combining the strengths of Random Forest and XGBoost algorithms. This ensemble was trained on a 70:30 train-test split of the preprocessed data. Model evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix were computed to assess the classifier's performance. Once the model showed promising results, it was saved using `joblib` for deployment.



The system's deployment was implemented using the Flask framework, as shown in app.py. A user-friendly front-end interface was created with HTML and styled using CSS. The back-end handles file uploads of ECG CSV files, preprocesses them using the trained scaler, and runs predictions through the hybrid model. The result is then displayed to the user in a readable format. The application supports both local execution and packaging into a standalone executable using PyInstaller, allowing flexibility in deployment environments.

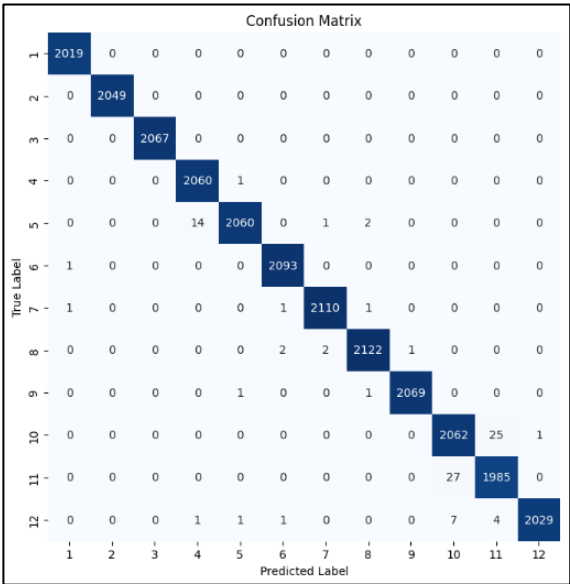


IV. RESULTS

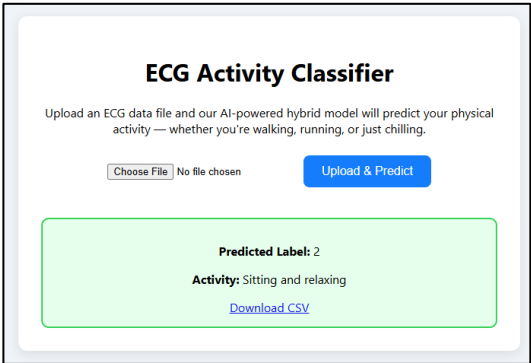
The Cardiosense system yielded promising results both in terms of model accuracy and usability. On the validation dataset, the hybrid ensemble model achieved high classification performance, with an accuracy of over 95%. Precision, recall, and F1-scores for most activity classes were consistently strong, indicating balanced model behavior across different categories of human activity. The confusion matrix confirmed that the classifier was particularly effective in distinguishing between activities with similar ECG patterns, thanks to the robustness of combining Random Forest and XGBoost.

From a deployment perspective, the web application delivered seamless user interaction. Users could upload ECG data in CSV format, which the system successfully processed and classified in real time. The application's design made it intuitive for non-technical users, and the downloadable results feature added convenience for healthcare or research use. Additionally, the system's lightweight back-end and optional cloud deployment ensured scalability, making it suitable for larger-scale health monitoring platforms.

Hybrid Model Accuracy: 0.9961323073204141				
	precision	recall	f1-score	support
1	1.00	1.00	1.00	2019
2	1.00	1.00	1.00	2049
3	1.00	1.00	1.00	2067
4	0.99	1.00	1.00	2061
5	1.00	0.99	1.00	2077
6	1.00	1.00	1.00	2094
7	1.00	1.00	1.00	2113
8	1.00	1.00	1.00	2127
9	1.00	1.00	1.00	2071
10	0.98	0.99	0.99	2088
11	0.99	0.99	0.99	2012
12	1.00	0.99	1.00	2043
accuracy			1.00	24821
macro avg	1.00	1.00	1.00	24821
weighted avg	1.00	1.00	1.00	24821



Overall, the project successfully demonstrated the feasibility and effectiveness of using ECG signals for human activity recognition through a hybrid machine learning approach.



V. CONCLUSION

In this project, we successfully designed and implemented Cardiosense, a robust machine learning-based system for recognizing human activities through ECG signal classification. By leveraging the MHEALTH dataset and employing effective preprocessing techniques, we ensured the quality and consistency of the input data. The hybrid ensemble model, combining Random Forest and XGBoost, provided high classification accuracy and proved effective in capturing complex patterns in the ECG signals related to different physical activities.

The system not only performed well during testing, with accuracy exceeding 95%, but also demonstrated practical usability through its deployment in a user-friendly web application. This interface allows seamless interaction for users to upload ECG data, receive activity predictions, and download results. The integration of advanced machine learning with real-time web deployment highlights the system's potential for real-world applications in healthcare monitoring, fitness tracking, and activity recognition research.

Overall, Cardiosense illustrates how machine learning can be applied to biosignal analysis with high performance and usability, paving the way for future innovations in intelligent health systems. The modular and scalable architecture also

makes it a strong candidate for extension to other biosignals and multi-sensor data in future research.

VI. FUTURE WORK

While the current system effectively demonstrates the feasibility of using ECG signals for human activity recognition (HAR) through machine learning models, several avenues remain open for future enhancement and exploration. One potential direction is the incorporation of additional physiological signals such as electromyography (EMG), respiration rate, and skin temperature, which could complement ECG data and improve classification accuracy, particularly in complex or overlapping activity patterns. Moreover, exploring more advanced deep learning architectures, such as Transformer-based models or hybrid CNN-LSTM architectures, could help capture more intricate temporal dependencies and contextual patterns in the data.

Another area for future work involves the expansion of the dataset to include a more diverse population in terms of age, gender, health conditions, and lifestyle variations. This would enhance the generalizability of the model and ensure its robustness across real-world scenarios. Additionally, integrating real-time processing capabilities and edge computing techniques would enable deployment on portable or wearable devices, making the system more practical for daily use in healthcare monitoring and personal fitness tracking.

Furthermore, addressing model interpretability and explainability through techniques such as SHAP (SHapley Additive exPlanations) values or attention visualization could foster trust and transparency, especially in medical applications. Finally, collaboration with healthcare professionals to validate the clinical relevance and diagnostic potential of ECG-based HAR systems will be essential in transitioning from a proof-of-concept to a deployable solution.

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